

What Data Centers Are Doing to Bees

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What Data Centers Are Doing to Bees Every chatbot query pulls a little more power from somewhere. Multiply that by billions of queries a day and you get a side effect almost nobody talks about: the data center boom is starting to interfere with bees. The clearest evidence points to electromagnetic fields. Data centers depend on substations, transformers, and high-voltage power lines to run, and in 2023 a study published in Science Advances found that EMF radiation from transmission towers puts real physiological stress on honeybees. Exposed bees showed altered gene expression in the systems that control navigation, foraging, and learning. In practice that meant they visited fewer flowers, and fewer flower visits means less pollination, which means fewer seeds and thinner plant diversity over time. The study's authors went so far as to suggest this could be a piece of the broader global pollination crisis. Part of why this matters is how bees actually find their way around. They rely on an internal magnetic compass, along with the position of the sun, to navigate. Flood that environment with man-made electromagnetic noise and you're essentially jamming their internal GPS. Heat is the second problem, and it works differently. Data centers use enormous amounts of electricity, much of it still coming from grids that lean on fossil fuels, so more data centers can mean more warming at a local level. Researchers at the University of Sydney modeled what this looks like for bees and came up with a striking number: somewhere between 3.5 and 14 million bees lost per year in Australia alone, driven by rising temperatures that increase mortality in eggs and larvae, reduce sperm quality in male bees, and give parasitic Varroa mites a bigger foothold. One planned data center cluster near Sydney is expected to draw as much electricity as two aluminum smelters combined. Recommended by LinkedIn None of this is really about a single cause. Water gets pulled for cooling. Land gets cleared for construction. Carbon output climbs. None of these things are aimed at bees specifically, but bees happen to be sensitive enough, and dependent enough on subtle environmental cues, that they absorb a lot of this fallout early. It's also worth being careful not to overstate where things stand. Researchers and beekeepers describe this as an emerging concern rather than a confirmed disaster, the science is still developing, and some of the more dramatic claims circulating online haven't been backed up by data. Still, the underlying mechanisms are real, and there's currently no federal standard limiting EMF exposure for wildlife, which leaves a real gap. The AI industry tends to present itself as weightless: you type something, you get an answer, and nothing physical seems to happen. But somewhere between the server rack and the power substation, that process has a physical footprint, and bees are one of the places it's starting to show up.